



The 2025 G7 Leaders' Summit will be held in Kananaskis, Alta., from June 15 to 17. As host of the G7, Canada has a chance to shape rules that will govern AI globally. (Shutterstock)

Canada has a chance to lead on AI policy and data governance at the 2025 G7 Leaders' Summit

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Artificial intelligence (AI) is rapidly transforming sectors from health care to climate science. But amid the global scramble to lead this technological revolution, one truth is becoming clearer: data, its platforms and its circulations, have become critical infrastructure. And Canada, poised to host this year's G7 Leaders Summit, has a rare opportunity to shape the rules that will govern AI globally.

Under the leadership of Prime Minister Mark Carney, the federal government elevated AI and digital innovation to a central pillar of national policy, and appointed Evan Solomon as minister of artificial intelligence and digital innovation. But ambition is not enough — Canada must now back its rhetoric with action that resonates at home and abroad.



Evan Solomon, minister of artificial intelligence and digital innovation, takes part in the cabinet swearing-in ceremony at Rideau Hall in Ottawa on May 13, 2025. THE CANADIAN PRESS/Christinne Muschi

Infrastructure intelligence

While AI headlines often focus on breakthroughs in generative models and robotics, the real engine of progress lies in less glamorous terrain: computing infrastructure and data systems.

Canada's proposal to build "next-generation data centres" is about creating the backbone for globally competitive and ethically governed AI. Without these facilities, modern AI systems cannot be trained, validated or deployed responsibly.

AI models — like those used in medicine for developing new drugs and health services, clean technologies such as clean energy and carbon-capture or materials science — require enormous computational power and massive datasets. That data must be structured, validated and — to the extent possible — open to those who can use it.



Ontario's electricity demand is expected to soar 75 per cent by 2050, in large part due to electric vehicle manufacturing and new data centres supporting AI. THE CANADIAN PRESS/Nathan Denette

Quality assurance

Our recent study underscores that the future of AI depends less on algorithmic cleverness and more on data quality and accessibility. Poorly labelled or fragmented datasets can introduce bias, reduce model performance or even endanger lives when used in health or safety applications.

Yet across many domains, useful data remains siloed and locked in proprietary formats, lacking documentation or inaccessible due to legal and technical barriers. This status quo serves monopolies, not society.

Canada holds the G7 presidency in 2025, and can provide leadership in data governance and AI innovation. A central priority should be to rally partners around a framework for ethical, accessible and well-designed datasets, especially in fields like health, climate science and materials research.

Tailored data

Our call for open data isn't one-size-fits-all. It must be tailored to the needs of specific sectors:

- Health-care AI requires anonymized patient data, genomic sequences, protein structure data, toxicology and carcinogen data, and drug response datasets.
- Climate AI needs long-term environmental records, satellite imagery, power and water use information and real-time emissions data.
- Materials science AI demands chemical interaction data, physical testing results, structural data and thermodynamic properties.

What binds these fields is a common challenge: ensuring data is ethically sourced, high-quality, and useable across borders and institutions. Canada's role should be to help build the platforms — digital, legal and diplomatic — that make this possible.

A G7 mandate

As host of the G7 in June, Canada can push for a transformative international commitment. At a minimum, this should include:

1. Common standards for open datasets, co-designed with input from AI developers, health professionals, climate researchers, materials scientists and legal experts.
2. Trusted data hubs, managed by public-private or non-profit entities, ensuring secure storage, privacy safeguards and public access.
3. Legal and diplomatic co-ordination, addressing cross-border data sharing, intellectual property constraints and ethical governance frameworks.

These steps would position the G7 — and Canada in particular — as a champion of AI that serves democratic values on top of commercial and geopolitical interests.

A logo for the G7 2025 Kananaskis meeting is pictured at a news conference in Banff, Alta. on May 22, 2025. THE CANADIAN PRESS/Jeff McIntosh

Canada's risks and opportunities

Canada is not starting from scratch. The country boasts leading AI research institutions, including the [Vector Institute](#) and [Mila](#), and has pioneered [open science partnerships](#) such as the [Montreal Neurological Institute's Tanenbaum Open Science Institute](#) and the [Toronto labs of the Structural Genomics Consortium](#).

Dataset platforms such as [AIRCHECK](#) (for AI-based chemical knowledge) and the [CACHE](#) competition (evaluating drug discovery models using open data), show how Canada is already putting together the building blocks of responsible AI. But the country risks squandering this advantage if it cannot scale these efforts or retain innovation domestically.

The stalled [Artificial Intelligence and Data Act](#) is a case in point. While the European Union moved forward with its [AI Act](#), the [General Data Protection Regulation](#) and the [European Health Data Space Regulation](#), Canada's legislative framework remains in flux.

Without clear domestic rules, and a proactive global agenda, Canada could end up as an incubator for innovations that [end up developed and applied elsewhere](#).

Global stakes

The AI race is not just about who builds the most powerful models. It's about who defines the technical, ethical and geopolitical standards that shape the digital future.

The G7 offers Canada a moment of strategic clarity. By investing in AI infrastructure and leading an international agenda on open, trustworthy AI, Canada can lead in shaping the rules.

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